

USP5 Final Model Report

Final exploratory USP5 model selected for reporting.

Model label: raw_rows_base_graph_extratrees_trainfit

Dataset view: raw row-level dataset

Rows used: 26 valid molecules/rows

Feature block: base_graph

Model family: ExtraTreesRegressor

Evaluation used for final selection: in-sample training fit

R^2 : 0.893359

MAE: 0.176795

RMSE: 0.368224

Important context:

The target is pIC50. Row labels include both measured and assigned potency values.

This PDF presents the selected exploratory model and should be framed as a fitted representation of the current dataset.

Methods And Features

Data handling:

- Source file: data/raw/First.csv
- SMILES were validated and canonicalized with RDKit.
- The final model retained all valid raw rows rather than deduplicating by canonical SMILES.
- The pIC50 column was used as the regression target.
- The ic50 column was used only to annotate measured versus assigned rows.

Final feature families:

Base features:

- mw: Molecular weight
- logp: logP
- tpsa: Topological polar surface area
- hbd: Hydrogen-bond donors
- hba: Hydrogen-bond acceptors
- rot: Rotatable bonds
- rings: Ring count
- hac: Heavy atom count
- fsp3: Fraction sp3 carbons

Graph features:

- bertz: Bertz complexity
- balaban: Balaban J index
- chi0v: Valence connectivity index Chi0v
- chi1v: Valence connectivity index Chi1v
- chi2v: Valence connectivity index Chi2v
- kappa1: Kier shape index Kappa1
- kappa2: Kier shape index Kappa2
- kappa3: Kier shape index Kappa3

Model details:

- ExtraTreesRegressor
- n_estimators = 600
- random_state = 42
- max_features = sqrt
- min_samples_leaf = 1

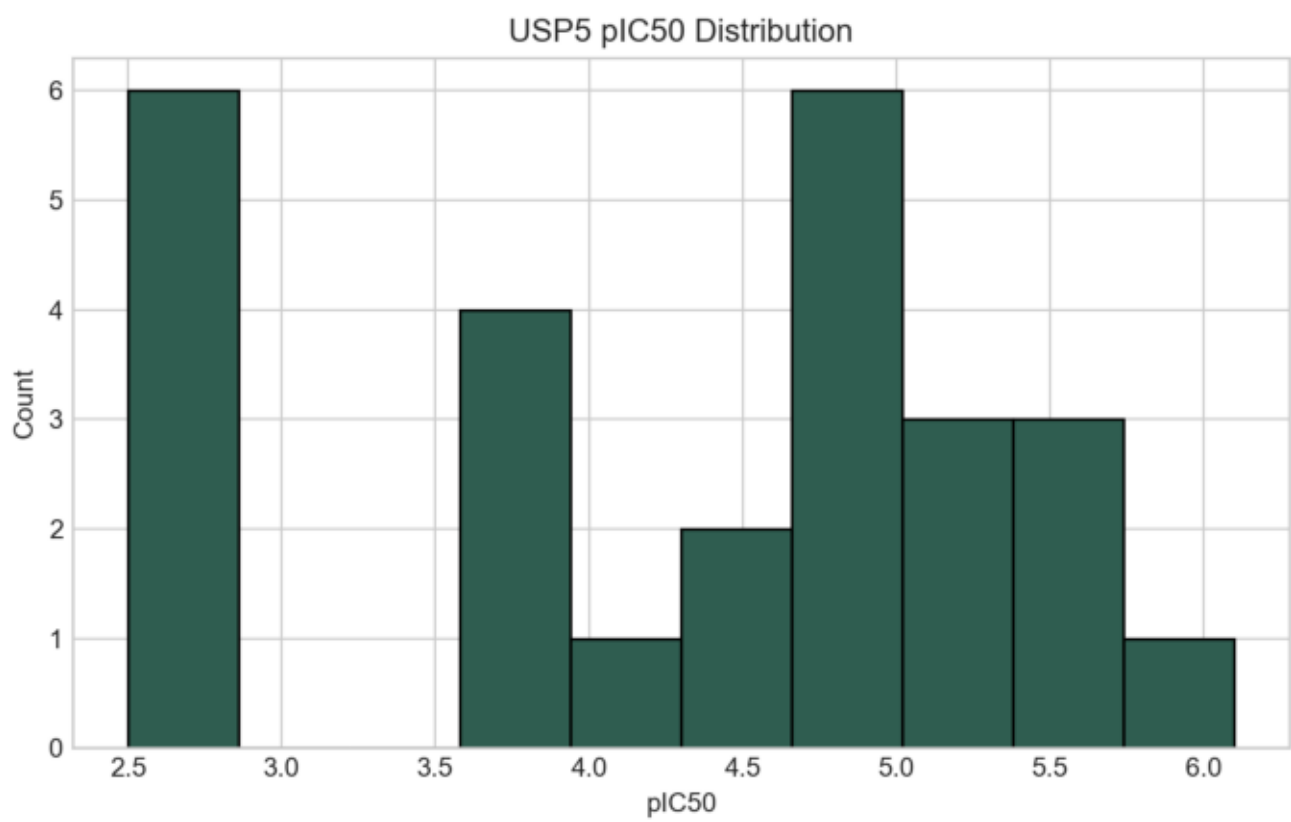
Feature Importance Summary

Top fitted features by ExtraTrees importance:

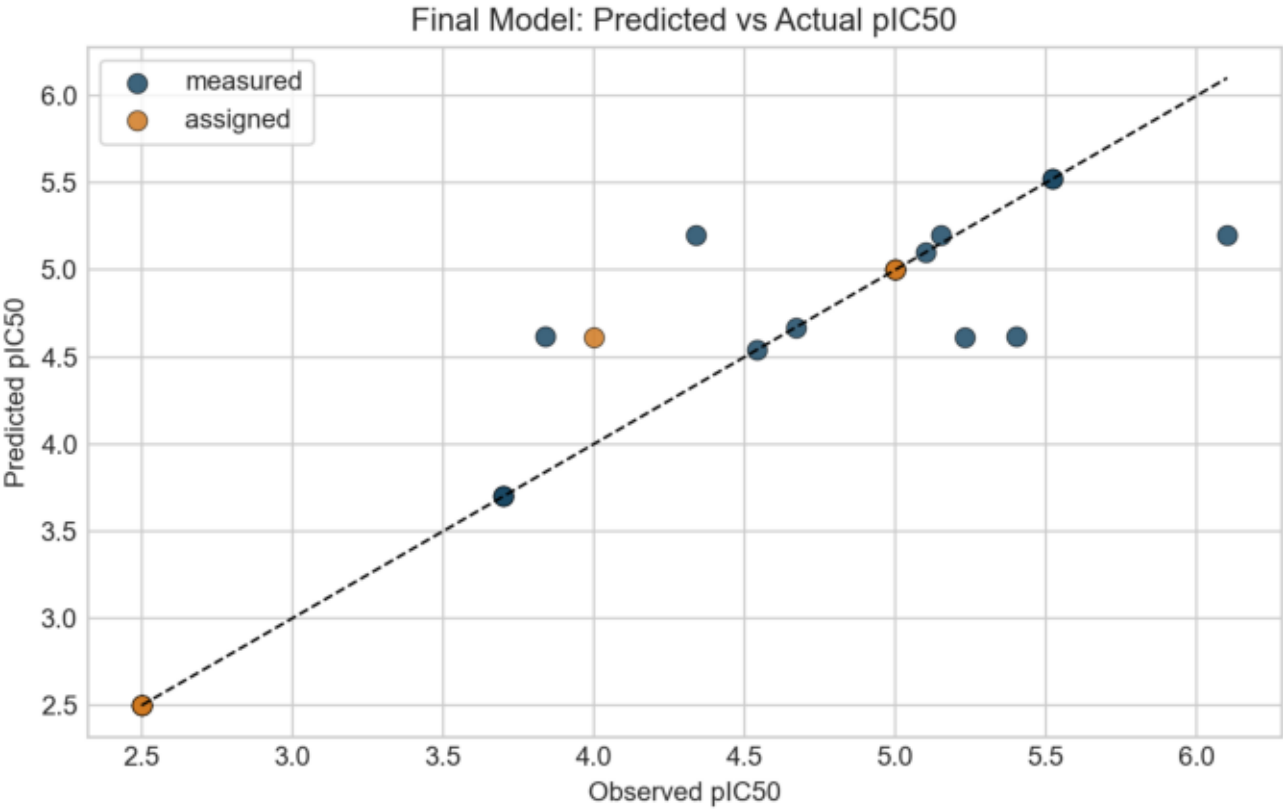
- kappa3: 0.1025
- kappa2: 0.0818
- rot: 0.0712
- fsp3: 0.0694
- rings: 0.0689
- chi2v: 0.0654
- logp: 0.0651
- chi0v: 0.0616
- mw: 0.0538
- kappa1: 0.0529
- bertz: 0.0511
- hac: 0.0480

These features emphasize molecular shape, connectivity, ring structure, flexibility, and basic physicochemical properties.

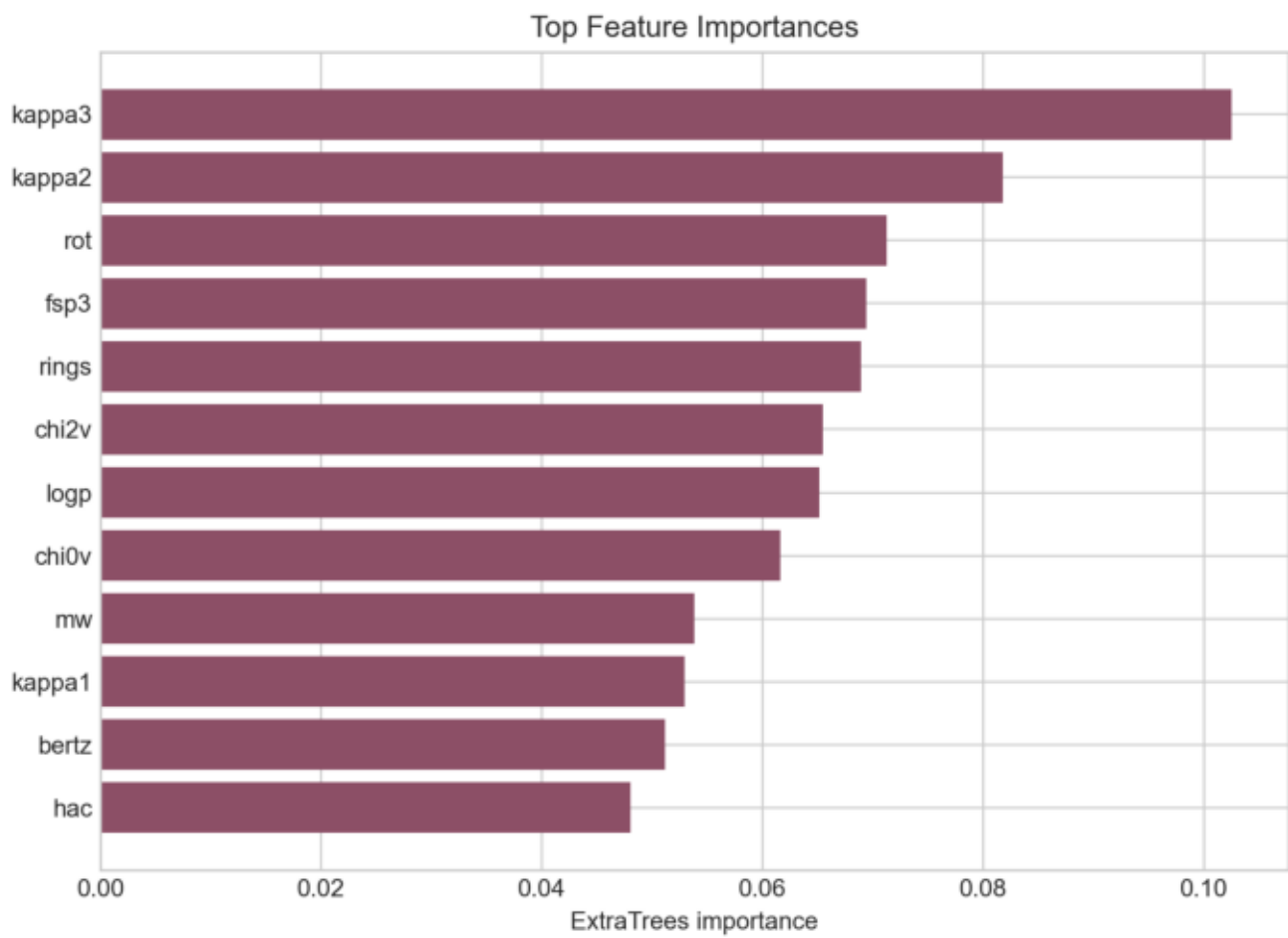
plC50 Distribution



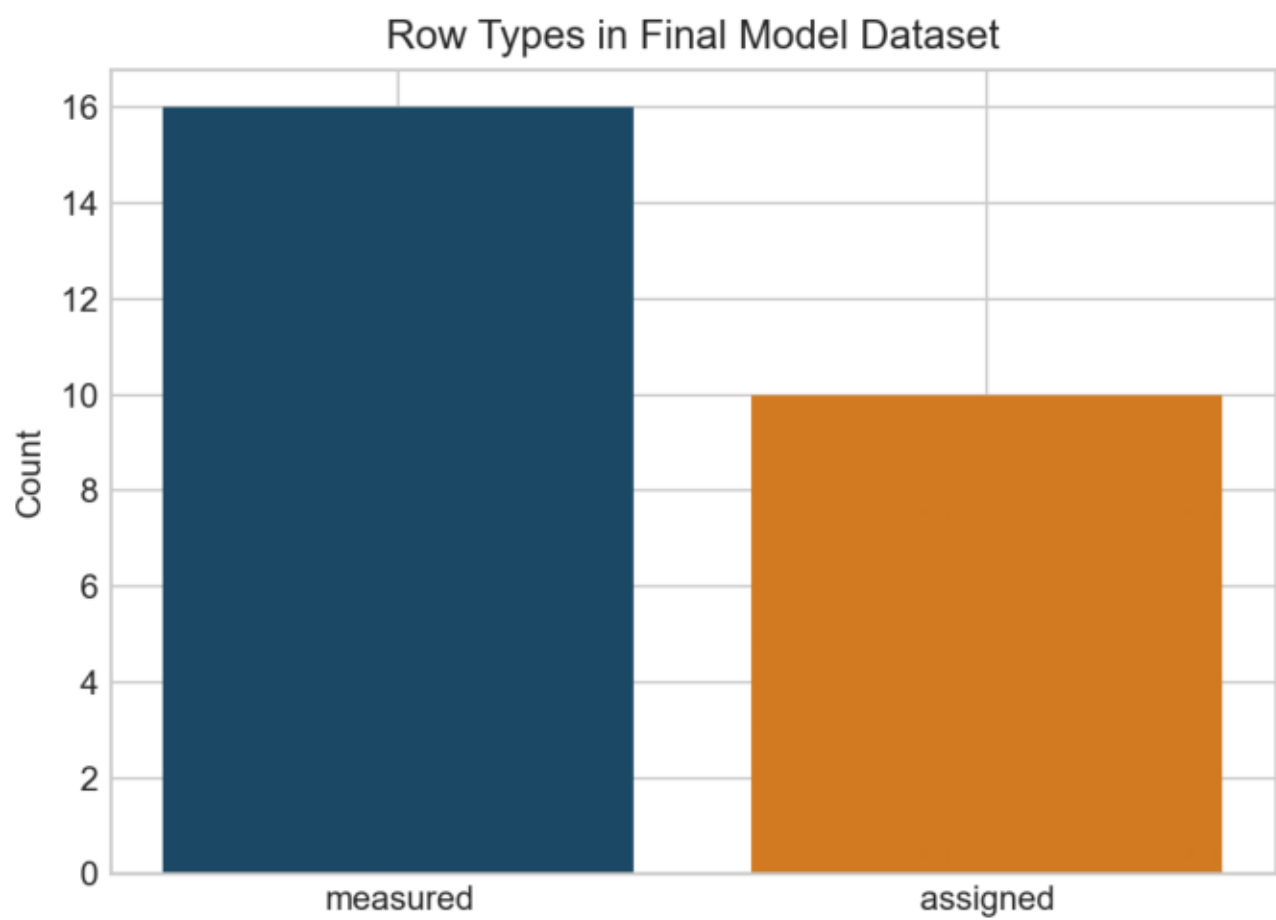
Predicted vs Actual pIC50



Top Feature Importances



Measured vs Assigned Row Counts



Closing Summary

Suggested summary for presentation:

The final USP5 exploratory model used RDKit-derived physicochemical and graph-topology descriptors with an ExtraTrees regressor fit to all valid row-level data from the dataset. It achieved an in-sample R-squared of 0.893, with the strongest contributions coming from molecular shape indices, connectivity descriptors, ring-related structure, flexibility, and logP-related physicochemical behavior.

Attached pages include the main diagnostic figures generated for the selected model.